Haotian Fu

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EDUCATION

Stony Brook University **Sep 2024 - June 2029 (Expected)**
Ph.D. in Computer Engineering. Advisor: Prof. Fan Ye, IEEE Fellow
Research Interests: Multimodal Sensing, Noninvasive Sensing, Smart Aging.
Shanghai Jiao Tong University (SJTU) **Sep 2020 - Aug 2024**
B.E. in Electrical and Computer Engineering (ECE) 2024 Outstanding Graduate of SJTU
Minors: Computer Science (CS), Data Science (DS)

RESEARCH PROJECTS

Graduate Research Assistant | Stony Brook University

Multimodal Dataset Collection, Curation, and Quality Assessment **Mar 2025 - present**

- Led data acquisition and curation for **113 long-duration sessions** of continuous activities of daily living (ADLs) in a four-room apartment sensing dataset, integrating 16 UWB sensors on 16 Raspberry Pis, 6 Kinect video/depth cameras, and 3 wearable watches/fingertips to support home-health monitoring.
- Developed and maintained an Android data-collection app and structured protocol with participant guidance and timestamped activity annotations, enabling repeatable multimodal recordings for downstream activity recognition, localization, and vital-sign monitoring research
- Implemented and tested automated data-quality and metadata workflows to check recording integrity, modality availability, and session-level consistency across heterogeneous sensing modalities.
- Mentored undergrad/master's students in data collection, metadata acquisition, quality assessment, and Android application development, coordinating contributions to ongoing dataset formulation.

Multi-Sensor Fusion for Robust Indoor Human Localization and Tracking **Nov 2024 - Feb 2026**

- Identified unreliable range estimation under multipath, occlusion, and clutter as a key obstacle to contactless, non-intrusive indoor human tracking during continuous ADLs, motivating robust monitoring in home and senior-care environments
- Developed ScoreKFT, a reliable and interpretable physics-informed range estimation and particle-filter-based multi-sensor fusion framework integrating 16 UWB sensors, achieving 0.958 m range RMSE and over 30% lower error than representative baselines.
- Built a DBSCAN-based ground-truth pipeline from Kinect video/depth data to reconstruct human positions and trajectories without prescribed walking paths.
- Evaluated continuous human localization and tracking across 12 long-duration ADL sessions using cross-modal depth-derived references in a fully furnished four-room apartment.

Cross-Population Generalization Evaluation for ADL Recognition (ADL-CLIP) **July 2026 - present**

- Reproduced a Transformer-based RF–text contrastive learning framework with PELT-based temporal segmentation and iterative representation refinement for continuous ADL detection.
- Conducted cross-population training and evaluation experiments for the ongoing journal extension, using Slurm-managed NVIDIA H200/B40 GPU clusters, and troubleshooting PyTorch/CUDA environments to assess model generalization across participant populations.

PUBLICATION

H. Fu, M. Liu, H. Han, Z. Xie, and F. Ye, "ScoreKFT: Robust Scoring-Based Range Estimation for Human Tracking in Realistic Home-Like Environments with UWB Radars," *2026 IEEE Global Communications Conference*, Accepted.

H. Liu, **H. Fu**, S. Yuan, C. Wu, Y. Luo and J. Li, "Adaptive Processing for Video Streaming with Energy Constraint: A Multi-Agent Reinforcement Learning Method," *2023 IEEE Global Communications Conference*, Kuala Lumpur, Malaysia, 2023, pp. 122-127, doi: 10.1109/GLOBECOM54140.2023.10437053.

TECHNICAL SKILLS

Programming: Python, C/C++, SQL, R, MATLAB
ML & Scientific Computing: PyTorch, CUDA, NumPy, SciPy, scikit-learn
Infrastructure & Tools: Git, Slurm, Docker, Linux, AWS (S3, Lambda)